

EXERCISE 2 · AFTER DECK 1

Tokens and Multi-Turn Conversations

Pressure-test the stateless-model mental model — tokens, context, history, cost.

6 questions · ~15–20 minutes · Worksheet only · answer key in separate PDF

WHAT TO DO

Six questions. About 15-20 minutes. Most of these are designed to catch the specific mistakes people make when they first try multi-turn. Trust the principles from Chapter 4.

Q1 OF 6 · CONCEPT

True or false: when you use a chat interface like ChatGPT or Claude.ai, the model **remembers** your previous messages across turns.

- A. True — the model has a built-in memory module that persists conversation state.
- B. False — the model is stateless; the UI re-sends the full conversation on every call.
- C. True for paid plans, false for free plans.
- D. True only within the same 24-hour window.

Your answer: _____

Q2 OF 6 · BUG SPOT

A learner tries to do multi-turn conversation. Their code below produces nonsense on turn 2 — the model doesn't seem to remember turn 1. Spot the bug.

```
conversation_history = []

# Turn 1
response_1 = client.converse(
    modelId=model_id,
    messages=[{"role": "user", "content": [{"text": "I like sci-fi."}]],
    inferenceConfig={"maxTokens": 200},
)

# Turn 2
response_2 = client.converse(
    modelId=model_id,
    messages=[{"role": "user", "content": [{"text": "Recommend something."}]],
    inferenceConfig={"maxTokens": 200},
)
```

Write your answer below:

Q3 OF 6 · CONCEPT

A model's context window is 200,000 tokens. Your system prompt is 500 tokens. After 30 turns, your conversation history has grown to 198,000 tokens. What happens on the next call?

- A. The model summarizes earlier turns automatically to free up space.
- B. The model truncates the oldest messages and continues silently.
- C. The API returns a `ValidationException` because input plus expected output exceeds the window.
- D. The model responds normally but charges you double tokens.

Your answer: _____

Q4 OF 6 · CONCEPT

Why is `maxTokens` described in the deck as your "hard ceiling against a runaway bill"?

- A. It caps how many input tokens the model can read.
- B. It caps how many output tokens the model can generate.
- C. It caps how many API calls you can make per minute.
- D. It caps how many models you can use in a single account.

Your answer: _____

Q5 OF 6 · BUG SPOT

Spot the conceptual error in this estimate:

"My system prompt is 1,000 tokens. I'm planning 20 turns. So the total tokens I'll send is $1,000 + 20 = 1,020$ tokens. Easy."

Write your answer below:

Q6 OF 6 · DESIGN SCENARIO

You're building a customer-facing chatbot. Conversations sometimes go past 50 turns. List **two different strategies** to keep cost and context-window usage under control, and name one tradeoff each strategy introduces.

Write your answer below:
